ELECTROMAGNETISM: ANSWERS

This is a closed-book examination. A formula sheet is provided

Question 1. [a) = 10] [b) = 5] [c) = 5] [d) = 8] [e) = 7] [f)= 5]

a) (calculated example)

i) The straightforward way is to show that $\nabla \times \mathbf{E} = 0$ [2]. Further [3 marks] are awarded for correct calculation of curl.

ii) By definition $\nabla\varphi = -\mathbf{E}$ [1]. Then

$$\frac{\partial\varphi}{\partial x} = -x \quad \frac{\partial\varphi}{\partial y} = -y \quad \frac{\partial\varphi}{\partial z} = -z \qquad [1]$$

The solutions are

$$\varphi = -\frac{x^2}{2} + f(y,z) \quad \varphi = -\frac{y^2}{2} + g(x,z) \quad \varphi = -\frac{z^2}{2} + h(x,y) \qquad [1]$$

The solutions can be reconciled, and $\varphi = -(x^2+y^2+z^2)/2$ [2]

b) (reasoning) Different valid examples can be given, such as (i) point charge or a any static charge distribution [1]; (ii) dc current flowing along an infinitely-long straight line or a closed loop [2]; (ii) antenna powered by an ac source [2].

c) (bookwork) These can be either stated directly or derived from the formula sheet knowing that the field inside a PEC is absent. The boundary conditions are $E_\tau = 0$ [2] and $\varepsilon_0 E_{n\,\text{out}} = \sigma$ [3].

d) (calculated example) The maximum value of mutual inductance will be achieved for total flux linkage between the two loops [2]. In this case, because the loops are identical, all self- and mutual- fluxes will be the same [2]. Denote the flux as Φ. Then

$$L = \frac{\Phi}{I} \quad [2]$$

and

$$M = \frac{\Phi}{I} \quad [2]$$

so that $M = L$.

e) (calculated example) A one-port device can be represented by a reactance X [1]. Then

$$S_{11} = \frac{jX - Z_0}{jX + Z_0} = -\exp\left(-2j\,\text{atan}\frac{X}{Z_0}\right) \quad [4]$$

As can be seen, $|S_{11}| = 1$, demonstrating that all incident power is reflected back (as is expected for a lossless passive device [2]).

Answers

d) (calculated example)

Far-field operation implies $r \gg \lambda$ [2]. For the frequency given $\lambda = c/f = 0.12$ m, and $r/\lambda = 2.45$, so that the communication does *not* take place in the far-field region of the antennas [3].

Question 2. [i) = 5], [ii) = 5], [iii) = 5], [iv) = 5], [b) = 10]

All questions are bookwork.

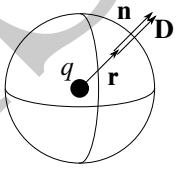
i) the atoms can be modelled as dipoles [2]. A simple model of an atom is an immobile nucleus surrounded by a spherical electron cloud [1]. In an external electric field, the centre of the electron cloud displaces relative to the nucleus [1]. Since the cloud is spherical, its electric field is the same as that of a point charge [1].

ii) free charges can, first, move under an electric field [1] and, second, they can exist in the absence of an electric field [1]. Bound charges, on the other hand, are unable to move [1]. They appear on the surface of a dielectric when the dielectric is placed into an external field [1] and they disappear once the dielectric is removed from the field [1].

iii) including free and bound charges into Maxwell's equations is inconvenient, given their different nature [2]. Therefore in macroscopic electromagnetic theory, bound charge is included into polarisation density [1], which in turn is included into vector **D** [2].

iv) when a dielectric is placed into an electric field, electric dipoles are induced inside the dielectric [1]. However, the charges of the neighbouring dipoles cancel each other [2] because the dielectric is overall electrically neutral [1] and uniform [1].

b) We place the charge at the origin of the coordinate system and look for $\mathbf{E}(\mathbf{r})$. We can make two observations based on symmetry. First, the magnitude of $\mathbf{E}$ is the same at equal distances from the charge [1]. Second, $\mathbf{E}$ is directed radially, which means $\mathbf{E} = E(r)\mathbf{e}_r$ [1]. We recall now that in a dielectric $\mathbf{D} = \varepsilon_d \varepsilon_0 \mathbf{E}$, so that $\mathbf{D}$ has the same properties as $\mathbf{E}$.



We now consider Maxwell's equation

$$\oiint_S \mathbf{D} \cdot ds = \iiint_V \rho dv \quad [1]$$

Surface S can be arbitrary, and we choose it to be the surface of a sphere of radius r centred at the point charge. Then the right-hand side is

$$\iiint_V \rho dv = q$$

The outward-pointing normal to the sphere is $\mathbf{n} = \mathbf{e}_r$, so that the left-hand side is transformed as

$$\oiint_S \mathbf{D} \cdot ds = \oiint_S \varepsilon_d \varepsilon_0 E(\mathbf{e}_r \cdot \mathbf{e}_r) ds = \oiint_S \varepsilon_d \varepsilon_0 E ds = \varepsilon_d \varepsilon_0 E \oiint_S ds = \varepsilon_d \varepsilon_0 E\, 4\pi r^2 \quad [3]$$

Answers

We used here the fact that $D = const$ on the sphere and took D out of the integral. Combining the left- and right-hand sides we get

$$\varepsilon_d \varepsilon_0 E\, 4\pi r^2 = q \quad [2]$$

and recalling that the direction of $\mathbf{E}$ is along $\mathbf{e}_r$ we finally get

$$\mathbf{E} = \frac{q}{4\pi\varepsilon_d\varepsilon_0 r^2}\mathbf{e}_r = \frac{q\mathbf{r}}{4\pi\varepsilon_d\varepsilon_0 r^3} \quad [2]$$

Question 3. [a) = 10], [b) = 5], [c) = 5], [d) = 10]

a) The short-circuited transmission line will have an impedance Z when referred to the ports. We can find the impedance from an expression given in the formula sheet. We have $\tan kd = \tan(2\pi\lambda/4\lambda) = \infty$ [2]. Then

$$Z = Z_0 \frac{0 + jZ_0\infty}{Z_0 + j0\infty} = \infty \quad [3]$$

Then at the frequency of interest, the short-circuited transmission line presents an open circuit to the ports, so that

$$S_{21} = 0 \quad [2] \qquad S_{11} = 1 \quad [3]$$

b) The device allows no transmission at f_0. As the frequency deviates from f_0, the impedance the shorted transmission line presents will become finite, leading to some transmission [2]. Therefore, the device is a notch filter around the frequency of interest [3].

c) The opposite function is that of a band-pass filter [1]. This function can be achieved by changing the termination of the transmission line from short to open circuit [3]. Open circuited transmission line will present zero impedance to the ports at f_0, leading to perfect transmission. Off this frequency, a finite impedance the line will present will lead to reduced transmission. [1]

d) We now need to consider the input impedance of the transmission line more carefully. Assuming $f = f_0 + \delta f$ where δf is small, we have

$$kd = \frac{2\pi f d}{c} = \frac{\pi}{2} + \frac{\pi(\delta f)}{2f_0} \quad [2]$$

so that

$$\tan kd = -\cot\frac{\pi(\delta f)}{2f_0} \approx -\frac{2f_0}{\pi\delta f} \quad [2]$$

Then

$$Z = jZ_0 \tan kd = -j\frac{2Z_0 f}{\pi\delta f} \quad [2]$$

The S-parameters can then we found using an expression from the formula sheet as

$$S_{11} = \frac{Z}{Z + 2Z_0} = \frac{1}{1 + j\pi\delta f/f_0} \quad [2]$$

and

$$S_{21} = \frac{Z_0}{Z + 2Z_0} = \frac{1}{1 - jf_0/(\pi\delta f)} \quad [2]$$

Answers